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Validity-Weighted Attenuation of Affective Noise in Open-Ended Teaching Feedback: An Interpretable ABSA Framework

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Abstract

Open-ended teaching evaluations are a widely used but underspecified data source for learning analytics, often carrying emotional content that is unrelated to instruction yet influences numerical ratings. Across 17,127 reviews from the anonymous, unmoderated platform RateMyProfessors.com, we find that non-pedagogical emotion ranks second only to instructional effectiveness as a driver of predicted ratings, and dominates the text of the lowest-rated reviews. We introduce Validity-Weighted Attenuation, an interpretable ABSA-based mechanism that mitigates the influence of construct-irrelevant affect on teaching-evaluation ratings under a definable pedagogical taxonomy. Each review is decomposed into clauses, classified against the taxonomy, and reduced to per-topic emotions; the modelled contribution of non-pedagogical affect is down-weighted in proportion to its measured density, with every adjustment traceable to textual evidence. Permutation controls support the mechanism's dependence on the original density assignment, and expert paired comparisons agreed with the relative magnitude of adjustment in 83% of pairs, exceeding the permutation null ($p = .001$); on close comparisons, agreement did not exceed the null. Adjustments are conservative: 60% of reviews are adjusted with an average absolute rating change of 0.20 points and the overall distribution shape preserved. Adjusted ratings are not validated estimates of teaching quality. We test whether an interpretable attenuation mechanism behaves consistently with a stated validity argument, offering initial method evidence for validity-aware interpretation of teaching feedback as a complement to raw ratings in learning analytics.

Notes for Practice

- **Users:** instructors and teaching/learning support staff interpreting open feedback.
- **Use:** inspect a review's original text, topics, and adjustment trail; use it to distinguish potentially actionable pedagogical feedback from other student experience signals.
- **Do not use:** automated ranking, promotion, tenure, discipline, or any high-stakes judgment about an instructor.
- **Interpretation:** an attenuated value is a model-based alternative view under a stated relevance taxonomy; it is not a validated estimate of teaching quality.

Keywords
learning analytics; teaching feedback; validity; construct-irrelevant variance; aspect-based sentiment analysis; interpretability

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1. Introduction

1.1 Open-Ended Teaching Feedback as a Learning-Analytics Trace

Open-ended teaching feedback is a widely available but analytically underspecified data source: nothing constrains how the content of accompanying text should shape interpretation of the rating, so any method that attempts to inform stakeholders of its validity must operate under cautious and auditable interpretation. Current methods, however, have been limited to primarily descriptive approaches. Sentiment analysis on Student Evaluations of Teaching (SET) is a growing area for computational insight into student experiences and institutional feedback. SET has historically guided consequential administrative decisions

(Haskell, 1997; Simpson & Siguaw, 2000; Irons et al., 2011). Despite this widespread usage, SET validity remains contested: ratings may reflect extraneous influences rather than teaching effectiveness (Langbein, 1994; Marsh & Roche, 1997). Situational factors, irrelevant content, and affective noise threaten validity (Wongsurawat, 2011; Schiekirka & Raupach, 2015), especially in informal instruments such as RateMyProfessors.com (RMP), which are fully open-ended, anonymized, and largely unmoderated. Interpreting SET results without accounting for contamination risks misinforming stakeholders and distorting faculty assessment (Haskell, 1997; Wongsurawat, 2011; Zumbach & Funke, 2014; Emery et al., 2003; Dowell & Neal, 1983).

Because a numerical rating collapses distinct sources of expression into a single value, the accompanying review text provides a trace through which those sources can be examined. This study treats open-ended teaching feedback as such a learning-analytics trace.

1.2 The Unresolved Methodological Gap

Despite advances in computational SET analysis, existing research remains largely descriptive. Sentiments are extracted, patterns identified, but intervention is rarely attempted. As a result, contamination documented in the literature continues to distort the numerical ratings that institutions and students rely on. We introduce **validity-weighted attenuation**, an interpretable ABSA method that proportionally attenuates the modelled contribution of affect in content classified as outside of a stated pedagogical taxonomy that stakeholders have the ability to control, while retaining the original review and a review-level audit trail. This methodology enables inspection of the framework's outputs and rating-adjustment process in a consequential application where important criteria cannot be fully specified (Doshi-Velez & Kim, 2017), and where analytical outputs must connect to pedagogical interpretation and action (Gašević et al., 2015). SET is an ideal domain: ratings are consequential, contamination is well-documented, and informal platforms like RMP represent a demanding stress test for validity-aware frameworks. Interpretability is essential given growing concerns about black-box systems in SET (Rybinski & Kopciuszewska, 2021; Zhuang et al., 2025), and findings may generalize to any open-ended feedback domain where numerical ratings are accompanied by unmoderated text.

Research Questions:

RQ1. How are topic-specific emotional signals associated with numerical ratings in open-ended teaching feedback?
RQ2. Can validity-weighted attenuation reduce the contribution of construct-irrelevant affect to numerical ratings while preserving relevant signals, under the taxonomy used in this study?
RQ3. Are the mechanism's coherence patterns stable under perturbation and consistent with expert judgment?

To answer our research questions, we develop and apply a five-stage NLP framework to 17,127 RMP student evaluations.

1.3 Contributions

The primary contributions of this research are:

- To our knowledge, the first computational validity-weighted attenuation framework that directly adjusts SET ratings by proportionally down-weighting emotional content in clauses classified outside of a stated pedagogical taxonomy, where each adjustment is traceable to specific topic-level emotions and content density.
- A theoretically grounded design bridging Messick's (1995) unified validity theory with classical reliability and robust estimation principles (Cronbach, 1951; Huber, 1964), translating established psychometric justifications into a concrete computational mechanism and laying groundwork for future validity-aware frameworks.
- Initial empirical proof-of-concept validation on a large held-out-instructor sample, with expert and robustness evidence.

This framework demonstrates that construct-irrelevant variance, as defined by the stated pedagogical taxonomy, can be attenuated in an interpretable manner, empowering stakeholders with additional information for cautious interpretation of teaching feedback (Tsai & Martinez-Maldonado, 2022). Content classified as *Miscellaneous* falls outside this study's taxonomy; it is not thereby unimportant to the student experience.

2. Background and Conceptual Framing

2.1 Validity, Construct-Irrelevant Variance, and Teaching Feedback

Messick (1995) identifies two primary threats to validity: construct underrepresentation and construct-irrelevant variance (CIV). His unified theory extends to any assessment in which responses are summarized into scores, a formulation that directly encompasses SET, with these threats particularly pronounced in informal, anonymous instruments such as RMP.

Marsh and Roche (1997) conclude that SET can be multidimensional, reliable, and relatively valid, but condition this on the content and coverage of items: “scores averaged across an ill-defined assortment of items offer no basis for knowing what is being measured" (p. 1187). Validity is thus a property of the content an instrument elicits; unconstrained, open-ended instruments offer no design guarantee on the composition of what they collect.

Wongsurawat (2011) identifies pedagogically irrelevant comment content as a potential source of measurement error in open-ended SET and argues for evaluating comment reliability. Comments with little to no correlation with average evaluations should be discounted, and administrators would benefit from metrics quantifying comment quality and reliability.

Across these studies, validity threats from construct-irrelevant content, including situational confounders (Schiekirka & Raupach, 2015) and students' emotional and relational states (Li et al., 2025), are documented in both formal and informal evaluation contexts. Recent psychometric work has demonstrated that CIV can be identified and mitigated in constructed-response assessments under controlled conditions (Zhai et al., 2021), but existing approaches either operate under such conditions, which do not extend to large-scale SET, or identify contamination without providing a computational mechanism to attenuate its influence on numerical ratings.

2.2 From Descriptive NLP to Validity-Oriented Interpretation

Prior SET research shows that aggregate sentiment can be associated with numerical ratings (Kechaou et al., 2011; Rajput et al., 2016; Iatrellis et al., 2024; Mihai, 2024), but aggregation obscures topic-specific emotional context. Deep-learning and ABSA (Aspect-Based Sentiment Analysis) techniques enable finer-grained representations of review content (Rybinski & Kopciuszewska, 2021; Shuqin & Raga, 2024); in SET, aspect-based approaches use this granularity to segment feedback and extract aspect-level sentiment (Bhowmik et al., 2023; Nikolic et al., 2020; Jazuli et al., 2025). Ren et al. (2022) extract and aggregate aspect sentiment, while Butt et al. (2025) compare BERT and LLM approaches for SET analysis.

Research on computational estimation of comment relevance shows that filtering by pedagogical value can improve prediction accuracy (Sorour et al., 2015; Louis & Nenkova, 2013; Xiong & Litman, 2011), but does not use relevance to alter affect's contribution to ratings. Neither line of work specifies how pedagogical relevance should modulate affect's contribution to rating interpretation.

2.3 Design Principle: Attenuation Rather Than Deletion

Addressing validity concerns within SET does not require excluding low-utility reviews. Classical reliability theory holds that measurement quality improves by reducing error variance rather than excluding observations (Cronbach, 1951). Robust estimation theory formalizes this principle, showing that down-weighting high-noise observations produces more stable and less biased estimates than exclusionary approaches (Huber, 1964). Within the SET domain, Marsh and Roche (1997) advocate for a conservative approach for bias mitigation, arguing that corrections should account for known sources of contamination while preserving students' overall evaluative intent. They further caution that corrections applied without regard to valid variance risk "throwing the validity baby out with the bias bathwater" (p. 1197), removing genuine signal alongside the contaminant. This supports a bounded, content-targeted adjustment rather than a global correction.

The attenuation mechanism itself is bounded and proportional, resulting in predictable changes to features. The taxonomy used in this RMP proof of concept is one generic decomposition of review content, and a different taxonomy would produce different adjustments. In formal applications, stakeholders could define or refine the categories in relation to course learning design and its documented pedagogical intent, so that the analytic lens reflects local priorities (Lockyer et al., 2013).

2.4 Research Gap

Despite advances in SET analysis, research offers limited computational support for operationalizing construct relevance to inform transparent, validity-oriented analysis of open-ended teaching feedback and its accompanying ratings. To our knowledge, no prior work links topic-level sentiment to an explicit model-estimated adjustment of accompanying ratings. Such an approach requires prioritizing transparency outputs that are auditable, contestable (Drachsler & Greller, 2016) and connected to pedagogical interpretation and action (Gašević et al., 2015).

3. Method

3.1 Study Context and Data

As a source of data, we use SET reviews from RateMyProfessors.com (RMP), a popular professor and course review website offering anonymous use by any student. As an informal SET system, RMP represents a deliberately noisy stress test for our framework's capacity to mitigate CIV, rather than a proxy for institutional SET. RMP lists a strict "no scraping" policy. Despite attempts to contact site administrators to request permission to source data directly from the site, no permission was granted; we therefore use the open-source dataset provided by He (2020), sourced from RMP. This dataset comprises more than 19,000 individual textual SET reviews and associated numerical ratings (1–5) directed towards hundreds of professors, alongside RMP meta-tags and demographic information measured by the original researchers. For our purposes, the meta-tags and demographic information are not used.

We excluded non-English reviews, reviews containing RMP's automatic censorship string "****" (whose application was prone to error), entries with no review text or text such as "No Comment", and reviews for professors with fewer than eight total reviews. After these exclusions, the final dataset contains 17,127 reviews.

Before analysis, reviews were cleaned (encoding repair via `ftfy`, lowercase normalization, abbreviation expansion, and punctuation standardization); stemming and lemmatization were withheld to preserve contextual information for downstream models.

3.2 Aspect-Term Categorization

Cleaned text is separated into individual clauses using `spaCy`'s `en_core_web_sm` model, which segments text based on punctuation while attempting to avoid splits on non-statement punctuation. When punctuation is sparse, two additional rules split text on commas and stop-words (but, so, yet), with aggressive stop-word separation used for reviews containing no punctuation.

The next stage in the processing pipeline assigns each review clause to one of three pedagogically relevant categories (*Instructional Effectiveness*, *Fairness & Grading*, and *Workload & Difficulty*) or to a *Miscellaneous* category. This task is formulated as an *aspect-term categorization* (ATC) problem.

Each pedagogical topic is defined using multiple short natural-language descriptions of common student expressions. These descriptions and each review clause are encoded using the pretrained sentence-embedding model `all-mpnet-base-v2`. For a clause c, cosine similarity is computed against each pedagogical topic, and the clause is assigned as follows:

$$\text{topic}(c) = \begin{cases} \arg\max\limits_{k\in\{\text{eff},\,\text{fair},\,\text{work}\}} s_{c,k}, & \text{if } \max_{k\in\{\text{eff},\,\text{fair},\,\text{work}\}} s_{c,k} > \tau, \\ \textit{Miscellaneous}, & \text{otherwise}, \end{cases} \tag{1}$$

where $s_{c,k}$ represents the cosine similarity between clause c and pedagogical topic k. We set $\tau = 0.25$, selected through validation on a development subset of manually inspected clauses to balance precision and coverage. By encoding both clause text and category descriptions as semantic vectors, the model captures meaning across the diverse phrasing and informal language common within RMP reviews.

Table 1 summarises the four-category taxonomy with representative descriptor phrases used to construct each topic embedding.

Table 1. Aspect-term categorization taxonomy and representative descriptor phrases.

Category	Representative Descriptors
Instructional Effectiveness	Clear and effective explanations of concepts; lectures are well-organized and easy to follow; helps students learn
Fairness & Grading	Grading feels fair, earned, and based on actual performance; tough but fair grader; adjusts grading to student's current ability
Workload & Difficulty	Heavy workload, lots of effort and time required; tests, exams, or quizzes are hard or demanding; easy class, light workload
Miscellaneous	Clauses below the similarity threshold τ for all pedagogical categories; includes personal remarks and social commentary

3.3 Pedagogical Density and Per-Clause Emotion Extraction

Using the categorized statements, density metrics are calculated for each review:

$$D_{\text{misc}} = \frac{\text{wordcount}_{\text{misc}}}{\text{total wordcount}}, \qquad D_k = \frac{\text{wordcount}_k}{1 + \text{wordcount}_k + \text{wordcount}_{\text{misc}}}, \qquad k \in \{\text{eff, fair, work}\}. \tag{2}$$

Here, D_{misc} represents the proportion of non-pedagogical (*Miscellaneous*) content; overall pedagogical density is therefore captured as $1 - D_{\text{misc}}$. The topic-specific density features ($D_{\text{effectiveness}}$, D_{fairness}, and D_{workload}) are also used as predictors in regression modelling, but not in the attenuation process.

Per-clause sentiment analysis is performed using `SamLowe/roberta-base-go_emotions`, a fine-tuned RoBERTa model trained on the GoEmotions dataset (Demszky et al., 2020). Each clause c is independently processed and represented by a 27-dimensional emotion vector $\mathbf{e}'_c$, where each component denotes the probability estimated by the model that the clause expresses an emotion label. The model initially produces the 28 GoEmotions labels; *Neutral* is removed because it represents the absence of strong emotional activation rather than an independent affective signal.

Using the previously obtained topic labels, clause-level emotion vectors within each review are aggregated at the topic level by computing the mean emotion vector over all clauses assigned to topic k:

$$\mathbf{E}_k = \frac{1}{|C_k|} \sum_{c \in C_k} \mathbf{e}'_c, \tag{3}$$

where C_k denotes the set of clauses in the review labelled with topic k. If no clauses are assigned to a topic, the corresponding $\mathbf{E}_k$ is imputed as a zero vector of length 27, preserving the fixed input dimensionality while indicating the absence of observed emotion for that topic.

3.4 Predicting Ratings from Topic Specific Emotions

Ratings are modelled as a function of emotions directed towards each topic and density metrics. For a single review, the input feature vector is represented as:

$$\mathbf{x}_{\text{review}} = \begin{bmatrix} \mathbf{E}_{\text{eff}},\ \mathbf{E}_{\text{fair}},\ \mathbf{E}_{\text{work}},\ \mathbf{E}_{\text{misc}}, \\ D_{\text{eff}},\ D_{\text{fair}},\ D_{\text{work}},\ D_{\text{misc}} \end{bmatrix} \in \mathbb{R}^{4\times 27+4} = \mathbb{R}^{112}. \tag{4}$$

Here, $\mathbf{E}_k \in \mathbb{R}^{27}$ is the aggregated emotion vector for topic k, and D_k is the corresponding density metric. For the complete dataset of $N = 17{,}127$ reviews, the feature matrix is $\mathbf{X} \in \mathbb{R}^{N\times 112}$.

To prevent data leakage and assess generalization to unseen professors, we employ an 80–20 professor-level grouped split. The CatBoost regressor is trained on 823 professors and evaluated on 206 held-out professors. Model-performance and coherence analyses use the held-out professors; the expert comparison uses a curated pair sample that spans both groups. This design produces more conservative performance estimates than random splitting while reflecting the challenge of generalizing across professors in larger-scale SET scenarios.

We employ CatBoost regression, an open-source gradient-boosted tree model that captures non-linear interactions between emotional signals and topics in the sparse review representation (Hancock & Khoshgoftaar, 2020; Prokhorenkova et al., 2018; Dorogush et al., 2018). The final configuration was selected through grouped 5-fold cross-validation (GroupKFold, grouped by professor ID) on the training professors: 1000 iterations, learning rate 0.02, depth 7, and L2 leaf regularization 6; all other hyperparameters were kept at their default values. MAE was used as both the loss function and evaluation metric because it measures average absolute deviation from ratings on the 1–5 scale and is resilient to occasional outliers.

3.5 Validity-Weighted Attenuation & Rating Adjustment

Raw ratings are adjusted to mitigate the impact of CIV while preserving pedagogical signals. Emotional features present in E_{misc} are down-weighted in proportion to D_{misc}, the density of *Miscellaneous* content in the overall review. The down-weighting factor ζ is calculated as:

$$\zeta = 1 - D_{\text{misc}}. \tag{5}$$

A review with no detected *Miscellaneous* content is unchanged ($\zeta = 1$), whereas a review composed entirely of such content receives zero weight for its *Miscellaneous* emotion features ($\zeta = 0$). Intermediate densities produce proportionate attenuation. The factor ζ implements the reliability and robust-estimation rationale for down-weighting high-noise observations rather than discarding them (Cronbach, 1951; Huber, 1964).

$$\hat{E}^{\text{weighted}}_{\text{misc}} = E_{\text{misc}} \cdot \zeta. \tag{6}$$

Applying the trained CatBoost model f_{CB}, held fixed across conditions, to the baseline and attenuated feature vectors produces:

$$\begin{aligned} \hat{y}^{\text{attenuated}}_i &= f_{\text{CB}}\left(\hat{\mathbf{x}}^{\text{weighted}}_i\right), \\ \hat{y}^{\text{baseline}}_i &= f_{\text{CB}}(\mathbf{x}_i), \qquad i = 1, \ldots, N. \end{aligned} \tag{7}$$

Adjusted ratings are defined as:

$$\hat{y}_i^{\text{adjusted}} = y_i^{\text{raw}} + \left(\hat{y}_i^{\text{attenuated}} - \hat{y}_i^{\text{baseline}}\right), \quad i = 1, \ldots, N. \tag{8}$$

The prediction difference $(\hat{y}_i^{\text{attenuated}} - \hat{y}_i^{\text{baseline}})$ functions as a correction Δ, quantifying the rating shift attributable to attenuating E_{misc}. Adding this correction to the raw rating changes only the model-estimated influence of *Miscellaneous* sentiment in proportion to D_{misc}.

3.5.1 Audit Trail and Worked Example

Table 2 presents an end-to-end trace of an individual review with high *Miscellaneous* density (D_{misc}=0.670), where non-pedagogical sentiment dominates, receiving a positive adjustment of +0.427.

Table 2. Worked Example, Review #14848.

Clause	Topic	Sim	Top 2 Emotions
dude is huge	*Miscellaneous*	0.030	admiration=0.044, approval=0.024
i mean huge	*Miscellaneous*	-0.010	approval=0.018, annoyance=0.015
is a rude guy	*Miscellaneous*	0.200	anger=0.567, annoyance=0.361
he knows alot about politics	Instr. Effectiveness	0.270	approval=0.293, admiration=0.031

Section 2: Baseline vs. Attenuated Feature Vector

Topic	Baseline Top 4 Emotions	Down-Weighted Top 4 Emotions
Instr. Effectiveness	approval=0.293, admiration=0.031, realization=0.016, optimism=0.004	approval=0.293, admiration=0.031, realization=0.016, optimism=0.004
Miscellaneous	anger=0.192, annoyance=0.129, admiration=0.016, approval=0.016	anger=0.063, annoyance=0.042, admiration=0.005, approval=0.005

Section 3: Validity-Weighted Attenuation

D_{misc}	ζ	Raw	Baseline	Attenuated	Δ	Adjusted
0.670	0.330	3.0	2.698	3.125	+0.427	**3.427**

Three of four clauses are Miscellaneous (D_{misc}=0.670), yielding ζ=0.330. Miscellaneous emotions are down-weighted by approximately 67%. The instructional effectiveness signal remains unchanged. The adjusted rating increases by 0.427.

4. Results

4.1 Corpus and Model Behaviour

4.1.1 ATC Topic Structure

Table 3 summarises topic frequencies after ATC. *Instructional Effectiveness* is the most frequent topic at both clause and review level, followed by *Miscellaneous* and *Workload*, with *Fairness* trailing substantially. The prominence of *Miscellaneous* clauses indicates that a substantial portion of student discourse falls outside pedagogical dimensions.

The rating distribution is negatively skewed, yet low-rating reviews show distinctive patterns: for ratings 1.0–2.0, *Miscellaneous* clauses are the most frequent topic, exceeding the next pedagogical category by approximately 27% in 1-star reviews.

Figure 1 shows the distribution of D_{misc} across all reviews. The distribution is heavily right-skewed (median 0.11, mean 0.22), with a spike at $D_{\text{misc}} = 1.0$ reflecting reviews composed entirely of non-pedagogical content, a subset receiving maximal attenuation.

Table 3. Clause and review-level topic frequencies.

Topic	Clause	Review	Avg./Review
Instructional Eff.	25,845	13,494	**1.92**
Miscellaneous	17,602	10,375	**1.70**
Workload	15,683	9,387	**1.67**
Fairness	4,258	3,567	**1.19**

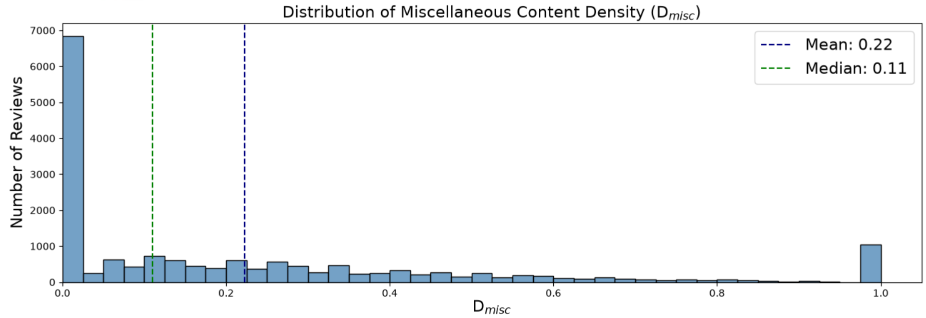

Figure 1. D_{misc} distribution across all reviews.

Together, these patterns show that pedagogical discourse centres on *Instructional Effectiveness*, that *Miscellaneous* content dominates the lowest-rated reviews, and that the right-skewed D_{misc} distribution motivates proportional rather than uniform attenuation.

4.1.2 Sentiment Extraction

RoBERTa-GoEmotions analysis reveals structured emotional patterns: positive emotions dominate high-rated clauses while negative emotions concentrate in low-rated clauses, though both extremes contain cross-valence signals. One-star clauses display a more evenly distributed profile than 5-star clauses, with notable positive-valence emotions, suggesting nuanced, multi-dimensional affect (Figure 2).

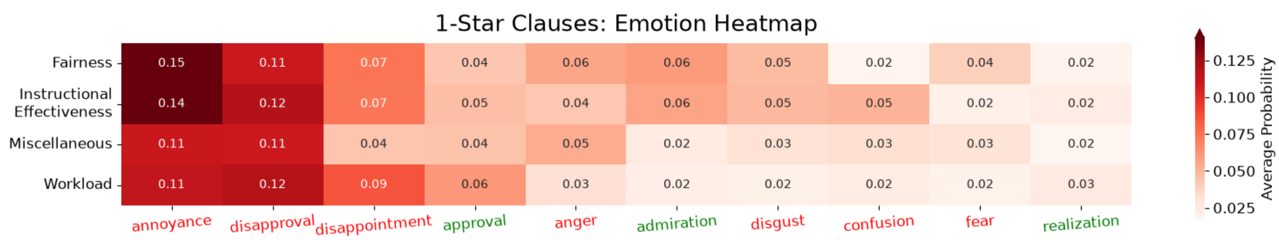

Figure 2. Average emotion probabilities by topic for one-star clauses.

4.1.3 ATC Validation

To assess whether the four-category taxonomy is conceptually distinct, two independent LLMs (GPT-5 mini and Grok 4.1) categorized 386 clauses, yielding Cohen's $\kappa = 0.7273$ ("substantial agreement"; Landis & Koch, 1977), supporting the clarity of category boundaries.

A senior faculty expert with more than 15 years of teaching and evaluation experience independently labelled the same 386 clauses, with multiple valid categories permitted per clause. Under a containment criterion (correct if the model's category appeared in the expert's label set), accuracy is 0.77. Restricting to clauses with exactly one expert label ($n = 287$) yields strict accuracy of 0.72; Table 4 reports per-category precision, recall, and F1.

Table 4. Single-label classification performance of ATC against expert annotations.

Category	Precision	Recall	F1	Support
Instr. Eff.	0.73	0.76	0.74	99
Fairness	0.57	0.76	0.65	17
Workload	0.63	0.78	0.69	54
Miscellaneous	0.82	0.66	0.73	117

	Precision	Recall	F1	Support
Macro Avg	0.68	0.74	0.70	287
Weighted Avg	0.74	0.72	0.72	287
Accuracy		0.72 (n = 287)		

These results indicate acceptable alignment for a proof-of-concept study, though accuracy leaves room for improvement, particularly for *Fairness* and *Workload* where support is limited.

4.1.4 Rating Regression Performance

The dataset is split 80–20 by professor to prevent leakage and better reflect the difficulty of generalizing to unseen instructors. Table 5 reports 5-fold GroupKFold CV and held-out performance; the held-out MAE of 0.6347 falls inside the fold range, confirming consistent generalization.

Table 5. CatBoost 5-fold CV and held-out performance.

	MAE	Spearman ρ	R^2
5-fold CV mean ±SD	0.6300 ±0.0081	0.7284 ±0.0152	0.6255 ±0.0088
Held-out professors	0.6347	0.7352	0.6168

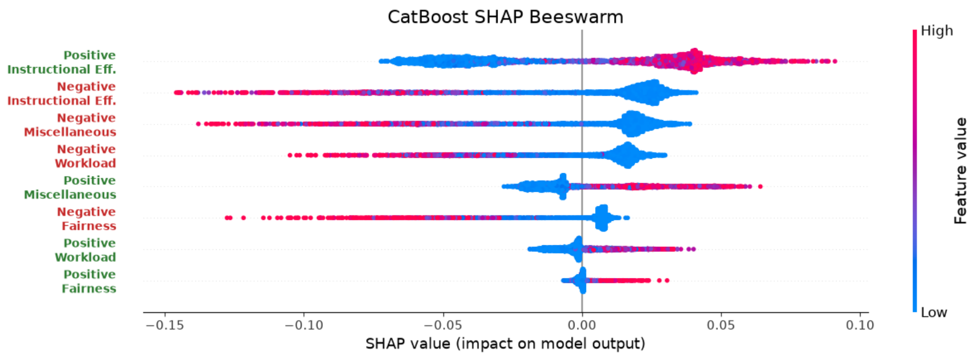

Figure 3. SHAP beeswarm plot for the eight polarity × topic categories on held-out professors. Point colour indicates feature value; horizontal position shows each feature's contribution to the predicted rating.

The final model demonstrates low absolute error, strong ordinal consistency, and substantial explained variance on held-out data, suggesting that it captures meaningful relationships between emotional content and student ratings.

SHAP values on held-out professors, grouped into eight polarity × topic categories, confirm directional coherence: high-intensity positive emotions increase predicted ratings while negative emotions decrease them, though low-intensity signals can reverse direction, indicating the model responds to the degree of expression rather than its mere presence (Figure 3).

4.2 What Attenuation Changes

We evaluate whether attenuation produces structured and interpretable changes, examining feature importance shifts, correlation changes, and the relationship between attenuation Δs and modulators (D_{misc}, E_{misc}).

4.2.1 Adjustment Outcomes

Over all 17,127 reviews, 10,375 (60.58%) were adjusted, with a mean $|\Delta|$ of 0.1964 and a mean signed Δ of 0.0440 (Wilcoxon signed-rank $W = 24{,}260{,}091$, $p < 0.001$); the professor-level distribution shape remained largely consistent.

The analyses that follow use the 206 held-out professors: 3,414 reviews for feature-importance analyses and 2,052 reviews containing *Miscellaneous* content for correlation and modulator analyses.

4.2.2 Mechanistic Coherence

Figure 4 and Table 6 summarise three coherence checks on held-out professors. SHAP-based feature importance shifts away from *Miscellaneous* sentiment and toward pedagogically relevant categories (Figure 4a). Pearson and Spearman correlations between baseline emotion features and ratings show the same pattern: *Miscellaneous* associations weaken while pedagogical associations are maintained or slightly strengthened (Figure 4b). These comparisons use the same emotion features before and after attenuation; only the rating changes.

Table 6. Correlation analysis: attenuation Δ versus modulators, computed on held-out professors ($n = 2{,}052$ reviews).

(a) Density modulator D_{misc}

Relationship	Pearson r	Spearman ρ
Δ^+ vs. D_{misc}	0.6872	0.6142
Δ^- vs. D_{misc}	−0.7077	−0.6001

(b) Intensity modulator E_{misc}

Relationship	Pearson r	Spearman ρ
Δ^+ vs. Neg. E_{misc}	0.4008	0.4327
Δ^- vs. Pos. E_{misc}	−0.2470	−0.1125

D_{misc} is the primary driver of attenuation magnitude (Table 6a): ratings increase or decrease in proportion to off-topic density, while correlations with E_{misc} intensity are weaker but directionally consistent. A Mann-Whitney U test confirms significantly larger absolute adjustments for high-noise ($D_{misc} > 0.5$, $n = 508$) than low-noise ($D_{misc} < 0.2$, $n = 690$) reviews ($U = 313{,}494$, $p < 0.001$, rank biserial $|r| = 0.789$). Together, these checks indicate that attenuation introduces structured and interpretable adjustments consistent with its design rationale.

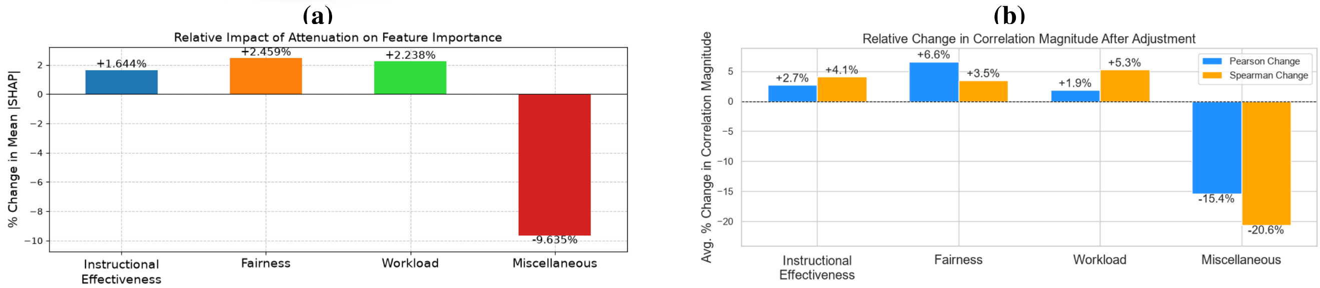

Figure 4. Mechanistic coherence checks after attenuation. (a) Change in SHAP-based feature importance (%), computed on held-out professors ($n = 3{,}414$ reviews). (b) Average change in correlation magnitude (%, Pearson and Spearman) after adjustment, held-out professors ($n = 2{,}052$). Bars show mean $(|\mathrm{corr}_{\mathrm{adjusted}}| - |\mathrm{corr}_{\mathrm{raw}}|)/|\mathrm{corr}_{\mathrm{raw}}|$ across polarity groups.

4.3 Initial Validity and Robustness Evidence

4.3.1 Expert Paired Comparisons

Three senior faculty experts with combined 75 years of experience in teaching, evaluation and supervision labelled paired reviews for relative validity selecting review 1, review 2, or "both valid." Experts were fully blind to any model outputs. 77 pairs were sampled from the full adjusted-review sample (43 coarse, 34 fine), with ATC assignments verified before inclusion. Coarse pairs were selected to have an absolute-adjustment difference $\big||\Delta_1| - |\Delta_2|\big| \geq 0.5$; fine pairs were selected with a difference below 0.5. At the operating point $s = 1$, these conditions partition the evaluable pairs.

Two matching votes established a majority. A model ranking agreed when the review receiving smaller $|\Delta|$ matched a directional majority; exact model ties selected the first displayed review. All retained pairs had three votes and a majority; one "both valid" majority counted as incorrect. Table 7 reports majority agreement, two-sided Wilson intervals, and nominal Fleiss' κ over all three response categories. Individual-expert fine agreement ranged from 58.8% to 70.6%. As a sensitivity check, excluding individual non-directional votes raised fine inter-expert agreement from nominal $\kappa = .469$ to generalized $\kappa = .634$; this describes agreement among retained directional votes and does not change the primary scoring.

Table 7. Three-expert paired comparisons. Agreement requires a matching directional majority; absent or non-directional majorities count as incorrect when deltas are available. Nominal κ uses all complete three-rater pairs in each condition (77 overall, 43 coarse, 34 fine), including non-directional responses.

Condition	Correct/n	Agreement	95% CI	κ
Overall	64/77	83.1%	[73.2%, 89.9%]	.658
Coarse	43/43	100%	[91.8%, 100%]	.819
Fine	21/34	61.8%	[45.0%, 76.1%]	.469

A post hoc diagnostic partitioned the 34 fine pairs by expert voting pattern: of 19 unanimous pairs, 16 agreed with the model ranking (84.2%); of 15 split-vote pairs, 5 agreed (33.3%). Ten of thirteen model–majority disagreements occurred on pairs where the experts themselves disagreed. Three disagreements persisted despite expert unanimity, at absolute-adjustment contrasts of 0.314, 0.005, and 0.026 points. We explore significance in the following section.

4.3.2 Permutation Control

For each of 1,000 permutations, we shuffled the positive D_{misc} values across the full dataset of 17,127 reviews and recomputed the attenuated ratings with the fitted model held fixed. The shuffled density supplied both a model feature and the attenuation input. Expert-pair agreement was then evaluated on the same 77 pairs. Modulator correlations were evaluated separately on the 2,052 held-out reviews with positive D_{misc}, using each review's original density as the reference. The primary expert test recomputed coarse/fine membership after each shuffle; a supplementary test kept the observed $s = 1$ membership fixed. Table 8 reports both specifications.

Overall expert agreement exceeded the shuffled-density benchmark ($p = .001$). When groups were recomputed, the coarse group averaged 10.51 pairs rather than the observed 43 and its membership changed; 179 shuffles reached 100% agreement ($p = .180$). Under shuffled density, the coarse group collapsed from 43 to a mean of 10.51 pairs, implying that the observed density assignment is responsible for the coarse and fine separation rather than the pair-selection protocol alone. On the original 43 coarse pairs, no shuffle matched the observed agreement ($p = .001$; null maximum .884). Neither fine analysis exceeded its null benchmark ($p = .378$ when groups were recomputed; $p = .321$ when they were fixed). Observed overall agreement exceeded the null mean by approximately four null standard deviations (0.198/0.049), while fine agreement fell within one null standard deviation of its null mean (0.019/0.054).

Table 8. Permutation control ($N = 1{,}000$). Expert rows report agreement; modulator rows report Pearson r. Null entries give the mean and one-sided p using $(k+1)/(N+1)$. Recomputed coarse groups contained a mean of 10.51 pairs (range 3–20), compared with 43 observed; fine groups averaged 66.49, compared with 34 observed. Fixed-bin subgroup analyses are supplementary; overall and modulator tests are shared.

Metric	Observed	Recomputed null (mean, p)	Fixed-bin null (mean, p)
Expert agreement (overall)	83.1%	63.3%, .001	Same test
Expert agreement (coarse)	100%	84.9%, .180	68.9%, .001
Expert agreement (fine)	61.8%	59.9%, .378	56.3%, .321
Δ^+ vs. D_{misc} (r)	.6872	.2327, .001	Same test
Δ^- vs. D_{misc} (r)	−.7077	−.1579, .001	Same test

No shuffle produced a positive modulator correlation as large, or a negative correlation as negative, as observed ($p = .001$ each). These results support dependence on the original density assignment under the specified perturbation, conditional on the fitted model and classifications.

4.3.3 Sensitivity to Attenuation Exponent

The linear attenuation function $\zeta = 1 - D^s_{\text{misc}}$ uses $s = 1.0$, operationalizing Huber (1964)'s principle that contaminated observations should be downweighted in proportion to their contamination. The sensitivity analysis below examines how the reported results vary with this choice. We evaluate across $s \in \{0.5, 0.75, 1.0, 1.25, 1.5\}$, symmetric around the operating point.

Table 9(a) reports selected delta comparisons: the two adjacent contrasts around $s = 1$ and the full tested span. Adjacent values yield near-identical adjustments (e.g., $s = 0.75$ vs. 1.0: $r = 0.995$, mean $|\Delta_a - \Delta_b| = 0.029$), and even the most distant pair ($s = 0.5$ vs. 1.5) retains $r = 0.964$, although rank agreement is lower ($\rho = 0.689$). The maximum single-review difference for that pair is 1.148, but this is one extreme case; the mean across all 2,052 reviews is 0.083.

Table 9. Robustness across attenuation exponents: (a) selected delta comparisons, with $d = |\Delta_a - \Delta_b|$; (b) expert majority agreement, with the same pairs and bins fixed at $s = 1$ ($n = 77$: 43 coarse, 34 fine; operating point in bold); (c) modulator correlations; (d) professor-bootstrap 95% confidence intervals. Panels (a), (c), and (d) use 2,052 held-out reviews with $D_{\text{misc}} > 0$; panel (b) uses the full expert sample.

(a) Pairwise delta agreement

s_a	s_b	r	ρ	Mean d	Max d
0.75	1.00	0.995	0.920	0.029	0.429
1.00	1.25	0.997	0.917	0.022	0.265
0.50	1.50	0.964	0.689	0.083	1.148

(b) Expert majority agreement

s	Overall	Coarse	Fine
0.50	88.3%	100%	73.5%
0.75	88.3%	100%	73.5%
1.00	**83.1%**	**100%**	**61.8%**
1.25	87.0%	100%	70.6%
1.50	87.0%	100%	70.6%

(c) Modulator correlations

s	$\Delta^+ r$	$\Delta^+ \rho$	$\Delta^- r$	$\Delta^- \rho$
0.50	0.677	0.505	−0.688	−0.502
0.75	0.684	0.576	−0.703	−0.542
1.00	0.687	0.614	−0.708	−0.600
1.25	0.686	0.656	−0.709	−0.646
1.50	0.677	0.680	−0.717	−0.694

(d) Bootstrap professor stability

Metric	Mean	SD	95% CI	
$\Delta^+ D_{\text{misc}}$ (r)	0.687	0.017	0.651	0.720
$\Delta^- D_{\text{misc}}$ (r)	−0.707	0.018	−0.741	−0.671
$\Delta^+ D_{\text{misc}}$ (ρ)	0.613	0.027	0.557	0.661
$\Delta^- D_{\text{misc}}$ (ρ)	−0.601	0.025	−0.649	−0.550

Table 9(c) shows that the modulator correlations change little across the tested exponents. Pearson correlations are nearly unchanged; the absolute Spearman correlations increase with s. The positive and negative relationships remain in the same directions throughout.

Table 9(b) holds expert-pair membership and coarse/fine bins fixed at $s = 1$, as in the primary expert analysis. Overall agreement ranged from 83.1% to 88.3% across the tested exponents ($n = 77$). Coarse agreement remained 100%, while fine agreement varied from 61.8% to 73.5% (21–25 of 34 pairs).

Table 9(d) reports 95% confidence intervals from 1,000 bootstrap resamples of held-out professors. Across resamples, the four correlations had SDs of 0.027 or less.

5. Discussion

5.1 A Validity-Oriented Analytic Primitive

RQ1 results show that *Instructional Effectiveness* is the primary driver of rating variation in the fitted model, dominating in both frequency and attribution. Non-pedagogical emotion ranks second only to *Instructional Effectiveness*, exceeding both *Fairness* and *Workload*, motivating an auditable method for examining its modelled contribution. Under the stated taxonomy, this is a potential source of construct-irrelevant variance, consistent with broader concerns about non-instructional influences on teaching evaluations (Marsh & Roche, 1997; Langbein, 1994; Schiekirka & Raupach, 2015; Dowell & Neal, 1983; Wongsurawat, 2011; Li et al., 2025). Rating extremes display different profiles: concentrated positive emotions in 5-star reviews versus diffuse negative emotions across all topics in 1-star reviews, with *Fairness* disproportionately represented in low ratings.

Addressing RQ2, held-out coherence checks show that attenuation redistributes feature attribution away from *Miscellaneous* sentiment (-9.6%) toward pedagogical categories ($+1.6\%$ to $+2.5\%$), and correlations shift in the intended direction. Adjustment magnitude tracks measured off-topic density in both directions, with weaker but directionally consistent emotion-intensity relationships. The average absolute adjustment of 0.1964 points and preserved professor-level distribution shape are consistent with the conservative character advocated by Marsh and Roche (1997).

For RQ3, the bootstrap and exponent checks show similar aggregate coherence patterns under the tested perturbations, while the permutation control shows that the modulator correlations depend on the original density assignment ($p = .001$). Although the operating point $s = 1$ yielded the lowest overall (83.1%) and fine (61.8%) expert agreement across tested exponents, all values fall within the respective 95% Wilson intervals ([73.2%, 89.9%] and [45.0%, 76.1%]); $s = 1$ was selected a priori for its direct proportionality to contamination density rather than optimised on expert agreement. The modulator correlations ($r = .687, -.708$) fall well above the permutation floor ($.233, -.158$) and the bootstrap CIs ([.651, .720] and [$-.741, -.671$]) confirm stability across professor samples and attenuation exponents. The Mann–Whitney rank-biserial ($|r| = .789$), computed on deliberately separated density groups ($D_{\text{misc}} < 0.2$ vs. $D_{\text{misc}} > 0.5$), indicates that approximately 89% of high-density reviews receive larger-magnitude adjustments than low-density reviews. In the expert comparison, the original coarse group exceeded its shuffled benchmark when membership was fixed; when membership was recomputed, the groups were smaller and changed across shuffles. Fine-condition agreement did not exceed the permutation null under either specification. The study cannot determine whether these disagreements reflect limits of the mechanism in close cases or differences between the experts' broader validity judgments and the taxonomy used here. Fine disagreements were concentrated in split-vote pairs, although three occurred despite expert unanimity.

The fine-condition agreement and permutation benchmarks provide a reproducible baseline for evaluating subsequent improvements to the taxonomy, density measure, or classification pipeline; small adjustment differences should not be interpreted as established validity orderings. These findings do not establish that the taxonomy or density measure is correct, nor that adjusted ratings are validated measures of teaching quality.

The framework addresses similar CIV concerns to those documented by Zhai et al. (2021) in controlled assessments, but in the considerably more challenging setting of informal, unmoderated SET reviews. This positions the work as initial method evidence for a validity-oriented direction in teaching-feedback analytics, rather than a ready intervention.

5.2 Implications for Responsible Interpretation of Teaching Feedback

At corpus scale, the framework can generate an auditable review queue that helps reviewers prioritize their inspection across large collections of feedback. Sorting reviews by the absolute difference between raw and adjusted ratings, $|\Delta|$, highlights cases in which content classified outside the stated taxonomy materially changes the modelled summary. These cases can be prioritized for inspection of the original review, raw and adjusted ratings, and topic-level audit trail, supporting pedagogical reflection informed by analytics, and prompting further investigation where appropriate (Tsai & Martinez-Maldonado, 2022; Gašević et al., 2015).

A second theoretical contribution is that the framework makes its evaluative taxonomy explicit and, in principle, configurable. Rather than treating pedagogical relevance as a fixed computational judgment, the categories can be defined around a stated purpose and local learning design (Lockyer et al., 2013). This makes visible a choice that is often implicit in feedback analytics: which dimensions should inform a particular interpretation of teaching feedback, and which should be treated separately.

At an aggregate level, topic-level emotional profiles may support exploratory identification of recurring concerns across courses or departments, provided they are interpreted in context rather than used as instructor rankings or performance measures.

5.3 Boundaries, Risks, and Non-Use

Additional information is not inherently safer to act on. The framework produces model-based adjustments under a stated pedagogical taxonomy. An attenuated rating is not a validated estimate of teaching quality; the framework decomposes a review into topic and emotion components, it does not summarize or replace the review itself. Any interpretation that bypasses the original text discards exactly the context the audit trail is designed to preserve.

Auditability alone does not make a deployment trustworthy. Drachsler and Greller's DELICATE framework treats transparency as one element of trusted learning analytics, alongside an explicit purpose, stakeholder involvement, data governance, and routes to contest harmful consequences (Drachsler & Greller, 2016). The mechanism can reproducibly compute review-level decompositions and adjustments; it must not autonomously rank, reward, penalize, or otherwise make consequential decisions about instructors or students. The present proof of concept does not authorize deployment in a consequential institutional or platform decision process.

The framework addresses one source of construct-irrelevant variance: the modelled contribution of affect in content classified outside the stated taxonomy. Other documented sources of bias in teaching evaluations, including gender, race, and popularity effects (Kim et al., 2025; Kopciuszewska & Rybinski, 2025), remain present in the data and in the adjusted ratings. A user who assumes that attenuation has produced a "cleaned" rating risks greater confidence in a result that still carries unaddressed contamination.

Responsible use requires understanding where the framework's auditability ends. The attenuation rule itself is fully transparent: $\zeta = 1 - D_{\text{misc}}$ is a fixed, monotonic function with no learned parameters. However, the components that produce its inputs are not equally interpretable. Topic assignment relies on sentence embeddings and cosine similarity thresholds, emotion classification uses a fine-tuned RoBERTa model, and the downstream rating model is a CatBoost ensemble. The audit trail shows *what* the framework did — which clauses were classified where, what emotions were detected, how the rating changed — but a user contesting an adjustment must recognize that the upstream classifications themselves are model outputs, not ground-truth labels. Engaging critically with the trail, rather than accepting it as a complete explanation, is part of using the framework responsibly.

6. Limitations and Future Work

This study has no independent external criterion against which to compare whether adjusted ratings support more defensible inferences about teaching than raw ratings. Mechanistic coherence, robustness checks, and expert paired- comparison agreement are indirect evidence; they do not demonstrate that adjusted ratings more accurately represent teaching quality. A stronger validity argument would require pre-specified, triangulated external evidence, for example, appropriately contextualized peer review, course-design evidence, and student-learning measures, evaluated on independent datasets. Such evidence is unavailable in the RMP corpus.

The exclusive use of anonymous, self-selected RMP reviews limits generalizability to formal SET, and the corpus's 5-star skew leaves fewer low-rating reviews for modelling. ATC relies on broad, single-label topics (strict accuracy 0.72), so multi-topic clauses may be misclassified, and the RoBERTa-GoEmotions model, trained on Reddit comments, may misinterpret sarcasm or informal language. Imperfect cleaning and clause segmentation can propagate errors downstream.

Attenuation outcomes depend on upstream classifications and the fitted model: high D_{misc} does not guarantee a large adjustment, and wholly *Miscellaneous* reviews ($D_{\text{misc}} = 1$) receive identical attenuated predictions regardless of emotional content. The word-count-based D_{misc} is only a proxy for relevance; future work could incorporate semantic similarity or other textual-quality measures.

Because the taxonomy may both omit pedagogically relevant concerns and miss other sources of construct-irrelevant variance, attenuation cannot be interpreted as a comprehensive correction for construct-irrelevant variance. Demographic information in the source dataset was not used, so this analysis cannot assess whether adjusted ratings retain gender or racial inequities.

The three-expert evaluation concerns relative review judgments on a small, curated pair set, rather than exact adjustment magnitudes. Agreement remains conditional on the panel and selection protocol, including verified ATC assignments; the expert sample spans training and held-out instructors. Wilson intervals condition on these experts and assume independent pairs. The unanimity breakdown is post hoc, and the 0.5 contrast cutoff defines the evaluated groups rather than an estimated discrimination threshold. Permutation interpretations remain conditional on the specified density shuffle and the model-based pair selection.

Future work should co-design and evaluate a formal-SET version of the framework with educators, students, and institutional stewards, using independent datasets and external teaching-quality criteria in line with the evaluative standards proposed by Ferguson and Clow (2017). Formal SET's structured prompts and standardized Likert scales may alter the distribution of *Miscellaneous* content, requiring recalibration. The formal-SET version should ground finer taxonomies in course learning design, potentially align them with SEEQ (Marsh, 1982), and use multi-label or soft assignment to distinguish, for example, procedural from interactional fairness and better represent multi-topic clauses. Before deployment can be considered, its purpose, access, retention, and contestation arrangements must be specified, and fairness audits should assess whether it amplifies inequities across instructor demographics and disciplines.

7. Conclusion

Open-ended teaching feedback is an increasingly common learning-analytics trace, yet computational approaches rarely move beyond description toward validity-informed intervention. This research presented validity-weighted attenuation, a method that produces adjusted ratings by attenuating the modelled contribution of affect in content classified outside a stated pedagogical taxonomy, offered as a complement to existing learning-analytics approaches for interpreting open-ended teaching feedback.

Its contributions are a novel attenuation mechanism with traceable, topic-level adjustments; a theoretically grounded design bridging psychometric principles (Messick, 1995; Cronbach, 1951; Huber, 1964) with a concrete computational mechanism; and initial proof-of-concept evidence from held-out instructors, expert comparisons, and robustness analyses.

Crucially, the framework preserves every original review and its full emotional profile; attenuation reweights the modelled contribution of affect to a summary rating. This design choice reflects a commitment to contestability: stakeholders can inspect and question the basis of any adjustment. As a proof of concept, the work shows how validity concerns can be made computationally explicit and auditable under a taxonomy stakeholders could define, providing a basis for future evaluation of validity-aware analytic tools.

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Ethical Considerations

All expert labelling was conducted under a REB exemption approved by the authors' institution, ensuring that annotation procedures adhered to ethical research standards.

Data & Code Availability

A public codebase for this framework is available online. Due to privacy restrictions, the original RMP dataset is not included; however, all fully anonymized expert-labelled data used for validation is provided.

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